

Characterization of a mould flux glass

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Abstract

A simplified mould flux glass composition used for the continuous casting of steel was synthesized and then characterized using X-ray powder diffraction (XRD) differential thermal analysis (DTA) and ¹⁹F and ²⁹Si magic angle spinning nuclear magnetic resonance spectroscopy (MAS-NMR). DTA showed the glass to have a low glass transition temperature and to crystallize readily at 600 °C. XRD of the heat-treated glass showed it to crystallize to cuspidine. ¹⁹F MAS-NMR showed the principal fluorine species to be F–Ca(*n*) with no evidence of Si–F or Al–F species. Fluoride ions therefore, complex calcium in this glass, rather than forming non-bridging fluorines. The network connectivity of the glass was calculated on this basis and found to be 2.07 this would be expected to correspond to a Q² Si species which was supported by the ²⁹Si data that gave a chemical shift of –78 ppm corresponding to Q² Si.

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1. Introduction

Mould flux glass compositions are used during the continuous casting of steel, they act as a lubricant between the water cooled mould and the steel. Their viscosity characteristics, heat transfer characteristics and the break temperature where there is an abrupt change in physical properties are all critical to the continuous casting process [1]. The break temperature generally corresponds to the crystallization of cuspidine (Ca₄Si₂O₇F₂). The types of glasses used as mould fluxes are summarized in Table 1. Mould fluxes are essentially based on calcium fluoro-alumino-silicate glasses. The alumina content is usually low and they generally contain magnesia and significant quantities of alkali metal oxides. The silica content is normally below 50 mol%.

Despite numerous attempts, successful modeling of the viscosity of mould flux compositions has not been achieved. Relatively little is known about the structure

of fluoro-silicate and fluoro-alumino-silicate glasses and in particular the role of fluorine in the glass structure is still controversial. Solid state ¹⁹F magic angle spinning nuclear magnetic resonance (MAS-NMR) is very useful for structural analysis of a variety of materials including glasses [4]. MAS probes with high spinning rates >10 kHz have contributed to reduction in line broadening and minimizing the need for multiple pulse techniques [5]. The ¹⁹F nucleus is 100% abundant and the most sensitive nuclei after hydrogen, with a spin = 1/2 and with a large degree of chemical shift anisotropy. The chemical shift of ¹⁹F ranges from +500 ppm to –500 ppm thus the chemical shift of ¹⁹F is very sensitive to its chemical environment. The observed ¹⁹F chemical shift has been shown to correlate with the fluorine–cation distance in alkali metal fluorides [6] and in alkali earth metal fluorides [7].

Several investigations have already been performed which examine F[–] sites in F-bearing silicates and aluminosilicates [4–10]. Stebbins and Zeng [10] have investigated the competition between cations of different field strengths with F[–] to form F–M(*n*). The results show

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Table 1

Composition of typical mole fluxes in mole fractions taken from [2] and [3]

	SiO ₂	CaO	CaF ₂	Al ₂ O ₃	Na ₂ O	MgO	Li ₂ O	NC1	NC2
LC2	0.36	0.24	0.14	0.04	0.10	0.04	0.09	0.82	2.07
MC2	0.32	0.33	0.16	0.02	0.06	0.03	0.08	−0.32	1.41
A	0.39	0.32	0.10	0.04	0.10	0.04	0.00	1.37	2.24
B2	0.36	0.26	0.25	0.03	0.09	0.01	0.00	0.13	2.44
C	0.43	0.30	0.16	0.03	0.02	0.06	0.00	1.27	2.60
D	0.45	0.31	0.13	0.04	0.02	0.06	0.00	1.71	2.67

Note small levels of species such as Fe and Mn have been neglected.

that F preferentially bonds to higher field-strength modifier cations. In another study by Youngman and Dejneka [11] carried out ¹⁹F MAS-NMR on aluminosilicate-lanthanum fluoride glasses. As well as LaF₃ species forming there was also evidence of Si–F and Al–F bonding. Zeng and Stebbins [5] have investigated the reduction of BO and formation of Al–F in aluminosilicates when F[−] is added and attempted to establish how this is dependent on the chemical composition of the glass. The proportion of non-bridging oxygens, NBOs, upon addition of F[−], in conjunction with the amount and type of modifying cations and the competition between modifying cations, Al and Si for F[−] ions, all influence the disruption of the silicate network. This in turn influences the macroscopic properties of the glass such as viscosity.

Hayashi et al. [12] have suggested that fluorine is dominantly coordinated with calcium in mould flux calcium fluoro-aluminosilicate glasses using XPS. Recently using MAS-NMR Wantanabe et al. [13] have also shown the presence of F–Ca(*n*) in CaO–CaF₂–SiO₂ glasses. However Kanehashi et al. [14] have suggested that some of the fluorine is present as Si–F bonds.

Recent studies [15–18] by the present authors on fluoro-alumino-silicate and fluorosilicate glasses have demonstrated that the structural role of fluorine in the glass closely matches that found in the crystal phase/s formed on heat treating the glass.

Table 2 summarizes the results of this study. For example in the glasses that crystallize to fluorapatite (Ca₅(PO₄)₃F) the fluorine is predominantly coordinated

by three close neighbor calcium ions, the same situation pertains to glasses which crystallize to fluorite (CaF₂) and to fluorcanasite (Na,K)₆Ca₅Si₁₂O₃₀F₄ and in both fluorite and fluorcanasite crystal structures the fluorine is surrounded by three or four calcium ions. A similar situation occurs in magnesium fluoro-silicate glasses with F–Mg(3) being found in the glass and in the crystal phase/s that form on heat treatment.

In mould fluxes the crystal phase that is formed is cuspidine, which also has fluorine surrounded by four calcium ions [19]. Our prediction based on our previous studies is therefore that F–Ca(4) will also exist in the mould flux glasses and will be the dominant fluorine species present.

Other possible fluorine species in mould flux compositions could include; Al–F–Ca(*n*), Al–F–Li(*n*), Al–F–Mg(*n*), Al–F–Na(*n*), F–Mg(*n*), F–Li(*n*), F–Na(*n*) and finally Si–F. The first four species are essentially a fluorine bonded to a four coordinate aluminum (Al(V)), which is then charge balanced by a cation. Given the low Al₂O₃ content concentration of mould fluxes these species are thought to be unlikely to be present in significant concentrations. Based on Zeng and Stebbins [5] work the higher field strength cations are more likely to complex F than lower field strength cations and therefore F–Ca(*n*) and F–Mg(*n*) are more likely to be present than F–Li(*n*) and F–Na(*n*). Given the relatively low concentration of magnesia in mould fluxes relative to calcia it is thought highly likely that the F–Ca(*n*) species will be the dominant species present.

Table 2

Summary of ¹⁹F MAS-NMR results from a wide variety of glasses

Glass code	Glass composition in mole percent	¹⁹ F chemical shift of glass (ppm)	Assignment	Crystal phase formed on heat treatment	Structural role of F	¹⁹ F chemical shift of crystal (ppm)
LG35	SiO ₂ (40.0)Al ₂ O ₃ (20.0)CaO(20.0)CaF ₂ (20.0)	−108 −150	F–Ca(<i>n</i>) Al–F–Ca(<i>n</i>)	CaF ₂	F–Ca(4)	−108
LG26	SiO ₂ (32.1)Al ₂ O ₃ (21.4)P ₂ O ₅ (10.7)CaO(21.4) CaF ₂ (14.3)	−108 −150	F–Ca(<i>n</i>) Al–F–Ca(<i>n</i>)	Ca ₅ (PO ₄) ₃ F	F–Ca(3)	−104
CAN4	SiO ₂ (61.6)CaO(19.2)CaF ₂ (10.3)Na ₂ O(3.8) K ₂ O(5.1)	−108?	F–Ca(<i>n</i>)	CaF ₂	F–Ca(4)	−108
LDIG24	SiO ₂ (42.2)Al ₂ O ₃ (7.7)P ₂ O ₅ (1.31)MgO(27.4) MgF ₂ (13.1)BaO(6.55)	−176	F–Ca(<i>n</i>) F–Mg(3)	(Na,K) ₆ Ca ₅ Si ₁₂ O ₃₀ F ₄ BaMg ₆ Si ₆ Al ₂ O ₁₈ F ₄	F–Ca(3) F–Mg(3)	−116 −176
F1	SiO ₂ (53.4)MgO(33.4)CF ₂ (6.6)Na ₂ O(3.3)K ₂ O(3.3)	−176	F–Mg(3)	KNaMg ₅ Si ₈ O ₂₂ F ₂	F–Mg(<i>n</i>)	−176

The presence of Si–F bonds with silicon complexing fluorine is thought to be unlikely in mould flux compositions, since there will be a high concentration of NBO non-bridging oxygens and the Si^{4+} ion will have a higher affinity for an O^{2-} ion than for an F^- ion. Thus it is thought that NBOs will be preferentially and almost exclusively found attached to Si and thus Si–F bonds will not exist in significant amounts.

Table 1 in addition to giving the composition of some mould fluxes also gives the calculated network connectivities as defined by Ray [20]. Network connectivity NC1 is the network connectivity calculated assuming fluorine forms non-bridging fluorines attached to Si and Al. NC2 is the network connectivity calculated assuming F is binding Ca and thus reducing the availability of calcium for forming NBOs on the Si. The values for NC1 are far too low to be credible and glasses would not be expected to form at such low NC values. The NC2 values are still low and are about 2.0, which corresponds to a silicon with two NBOs linked via oxygen to two more silicons. (N.B. these Si could be replaced by Al) (Q^2). A glass with $\text{NC} = 2.0$ is likely to consist of a linear silicate chain of infinite molecular weight. At a $\text{NC} = 2.0$ the glass composition will be very sensitive to composition, reducing the network modifier content by a very small amount converts the glass from linear chain structure to a three dimensional network and would be expected to result in a large viscosity increase. In contrast increasing the network modifier content by a small amount would reduce the molar mass of the silicate chains and result in a marked viscosity decrease. Thus the viscosity of mould fluxes will be very sensitive to small compositional changes.

One surprising aspect of having F–Ca(n) present is that incorporating fluorine is reducing the NBO content of the glass, yet fluorine is known to reduce the glass viscosity. This can be explained by the fact that a Ca^{2+} ion can effectively ionically bridge two NBOs and thus is not

as disruptive to the glass network as two Na^+ ions. Incorporation of fluorine may effectively convert a Ca^{2+} species to a CaF^+ species, which may then act more like an alkali metal cation (Fig. 1).

2. Experimental

The mould flux glass LC2 given in Table 1 was made by melting brazilian grade quartz (99.99% SiO_2) Al_2O_3 , CaCO_3 , CaF_2 , $\text{MgONa}_2\text{CO}_3$ and Li_2CO_3 in a platinum crucible at 1350°C for 1 h. A batch size of approximately 100 g was used. The resulting melt was quenched into water. The frit was dried immediately and ground in a vibratory mill to give a fine powder $<150\mu\text{m}$ which was used in the subsequent analysis.

Synthetic cuspidine was made by melting 200 g of SiO_2 , CaCO_3 and CaF_2 in stoichiometric proportions in a platinum crucible at 1450°C , 40°C above the melting point of cuspidine for 30 min. The resulting melt was poured on to a stainless steel plate and allowed to cool slowly to room temperature.

A combined differential thermal analyzer–thermal gravimetric analyser (DTA–TGA) (Stanton Redcroft STA 1640 (Rheometric Scientific, Epson, England) was used to characterize this glass. A heating rate of 10°Cmin^{-1} was used with a flowing argon atmosphere with matched platinum crucibles. X-ray powder diffraction (XRD) was carried out on the glass powder and on the powder heated at 10°Cmin^{-1} to 550 and 600°C , which corresponds approximately with the temperature of the crystallization exotherm observed on the DTA trace. The synthetic cuspidine was ground to a powder and examined by XRD.

Both the original glass and the glass heat treated glass were examined by both ^{19}F and ^{29}Si . The ^{19}F measurements were conducted at a resonance frequency of 188.29 MHz and the ^{29}Si measurements at 39.77 MHz using an FT-NMR spectrometer (DSX-200, Bruker, Germany). The spinning rate of the samples at the magic angle were 10–5 kHz for ^{19}F and 5 kHz for ^{29}Si . The recycle time was 120 s for ^{19}F and 2.0 s for ^{29}Si . The ^{19}F spectra were recorded using an anti ring pulse sequence to suppress the probe background. The reference materials for the chemical shift were CaF_2 taken as -108ppm and tetramethylsilane.

3. Results and discussion

The glass frit was optically clear and X-ray powder diffraction showed it to be amorphous. The X-ray powder diffraction pattern is shown in Fig. 2.

The cuspidine melt crystallized on cooling to give long acicular crystals of 1–6 mm in length. XRD of the synthetic cuspidine showed it to be approximately

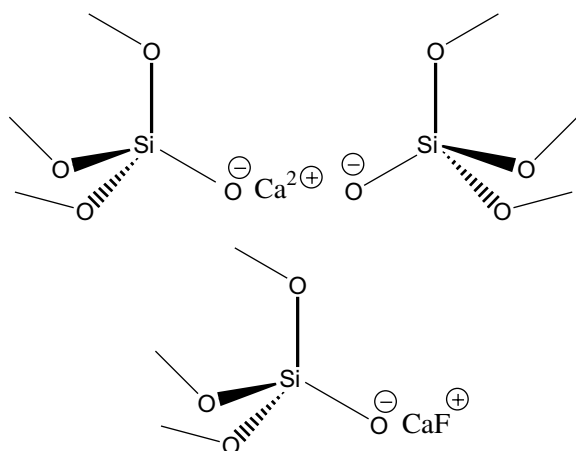


Fig. 1. Schematic of possible calcium species.

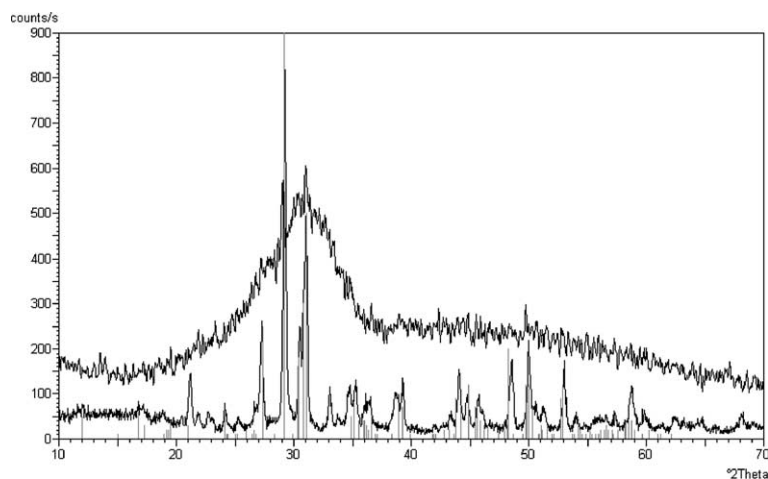


Fig. 2. XRD patterns of the LC2 glass (top) and the LC2 glass heat treated to 600°C (bottom). The lines indicate the match to cuspidine.

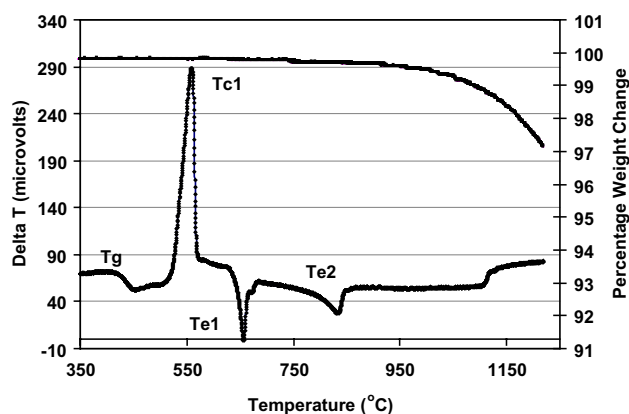


Fig. 3. Combined DTA–TGA results for the LC2 glass.

100% cuspidine. The combined DTA–TGA results are shown in Fig. 3. There is a sharp glass transition temperature at 421°C followed by a very sharp crystallization exotherm at 562°C. These are followed by two endotherms at 658°C and 838°C. The TGA curve shows a weight reduction starting at 500°C with a 2.5% weight reduction by 1250°C. This weight reduction is most likely to correspond to the loss of silicon tetrafluoride. Similar weight losses have been observed in a wide range of fluoro-alumino-silicate glasses [20,21].

The glass heat treated to 550°C had undergone only very slight crystallization. The glass heat treated to 600°C has crystallized to a large volume fraction of cuspidine (Fig. 2). This mould flux composition [2] like most mould flux compositions crystallizes readily to cuspidine. Fig. 4 shows the ^{19}F MAS-NMR spectra for the glass and the heat treated glasses.

The glass shows a relatively broad peak at -108 ppm corresponding to $\text{F-Ca}(n)$. The ^{19}F spectra of the synthetic cuspidine is also shown in Fig. 4 and exhibits two sharp peaks at -100 and -105 ppm. The doublet

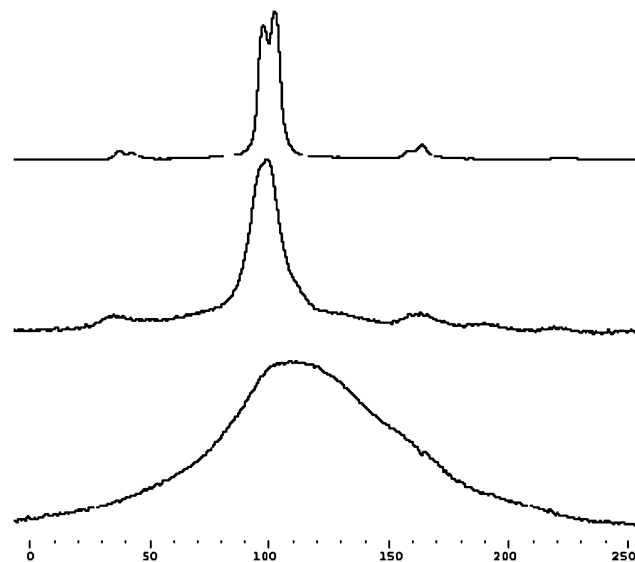


Fig. 4. ^{19}F MAS-NMR spectra for the LC2 glass (bottom) the LC2 glass heat treated to 550°C, and heat treated to 600°C and synthetic cuspidine (top).

peaks either side of the main peaks of the cuspidine are spinning side bands. Watanabe et al. [22] also found cuspidine to exhibit two sharp peaks in their synthetic cuspidine with chemical shifts of -99 and -104 ppm. The two peaks correspond to two slightly different $\text{F-Ca}(n)$ environments in the crystal structure [19]. The heat treated glass showed a single sharp peak at -100 ppm corresponding to the two $\text{F-Ca}(n)$ sites in cuspidine. The fact that only one peak was observed in the cuspidine from the heat treated glass is probably a result of some slight disordering of the $\text{F-Ca}(n)$ sites in the crystal, which results in broadening of the $\text{F-Ca}(n)$ peaks. There is relatively little fluorine remaining in the residual glass phase after heat treatment at 600°C. The ^{19}F MAS-NMR results therefore support the

hypothesis that fluorine is present largely as F–Ca(*n*) in the glass and that fluorine is complexing calcium. There is a slight asymmetry to the spectra for the glass which may indicate that small amounts of other species are present including F–Mg(*n*) at about –176 ppm and Si–F species such as Si–F–Ca(*n*) and Si–F–Na(*n*) in the range –116 to –130 ppm. The peaks either side of the main peak of the heat treated glass are spinning side bands. However the spinning side band at –225 ppm is enhanced and probably corresponds to a small amount of F–Na(*n*).

On the basis of fluorine complexing calcium in this glass the glass has a calculated network connectivity of 2.07, which corresponds to Si in a Q² environment. The ²⁹Si spectra of the glass and the heat treated glass are shown in Fig. 5. Engelhardt et al. [23] described that the chemical shift of Si in aluminosilicate glasses in terms of both the number of bridging oxygen (*m*) per SiO₄ unit and number of aluminum (*n*) connected by oxygen with the SiO₄ unit. When a structural unit of the aluminosilicate glass network is expressed as Si(OSi)_{*m*–*n*}(OAl)_{*n*}(O[–])_{4–*m*}, (Q^{*m*}(*n*Al), 4 ≥ *m* ≥ *n* ≥ 0), the chemical shift increases with decreasing *m* or increasing *n*. Given the low alumina content of these glasses the influence of *n* will be negligible and the chemical shift would be expected to depend almost entirely on *m*. For a glass with *m* = 2, i.e. a Q² silicate structure a chemical shift in the range –76 to 80 ppm would be expected and indeed this is found for this glass that exhibits a broad peak with a chemical shift of –78 ppm. The crystallized glass containing cuspidine has a two overlapping peaks at –70 and –72 ppm, which would correlate with the two slightly different F–Ca(*n*) sites found in synthetic cuspidine [19,22]. The cuspidine would be expected to have a Q¹ structure with a chemical shift about –70 ppm. The fact that two peaks are present suggests that the two Si atoms in cuspidine may have slightly different chemical environments. The residual glass phase would be expected to be slightly more cross-

linked following the crystallization of cuspidine and the Q structure would be expected to move from Q² towards Q³ and the chemical shift observed to become more negative and this is indeed the case with a chemical shift of –82 ppm. However the peak at –82 ppm is surprisingly sharp for a glass phase and there are also a number of weaker peaks present that are difficult to interpret.

4. Conclusions

In the mould flux composition studied fluorine is present principally as F–Ca(*n*) species with no evidence of either Si–F or Al–F species. Thus there is no evidence for non-bridging fluorines. Network connectivity calculations assuming fluorine complexes calcium give rise to a network connectivity of about two for mould flux compositions. The viscosity of such compositions will be very sensitive to chemical composition.

Further MAS-NMR studies including ²⁹Si, ²⁷Al and ¹⁷O on mould flux type compositions where the components in the composition are systematically varied are planned for the future. Such studies will be valuable in designing and producing new mould flux compositions that hopefully will lead to improved efficiency in the continuous casting of steel.

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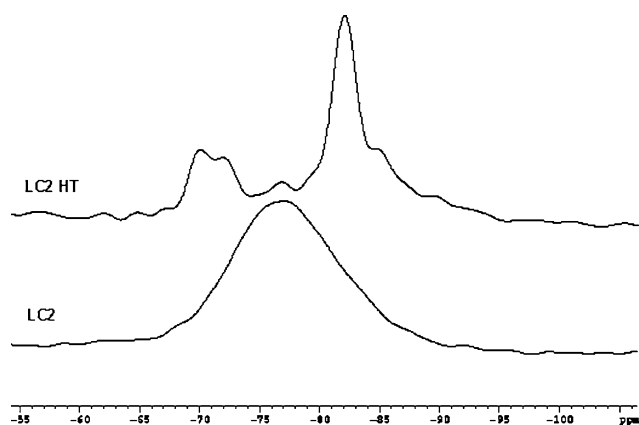


Fig. 5. ²⁹Si MAS-NMR spectra for the LC2 mould flux composition before and after heat treatment at 600 °C.

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